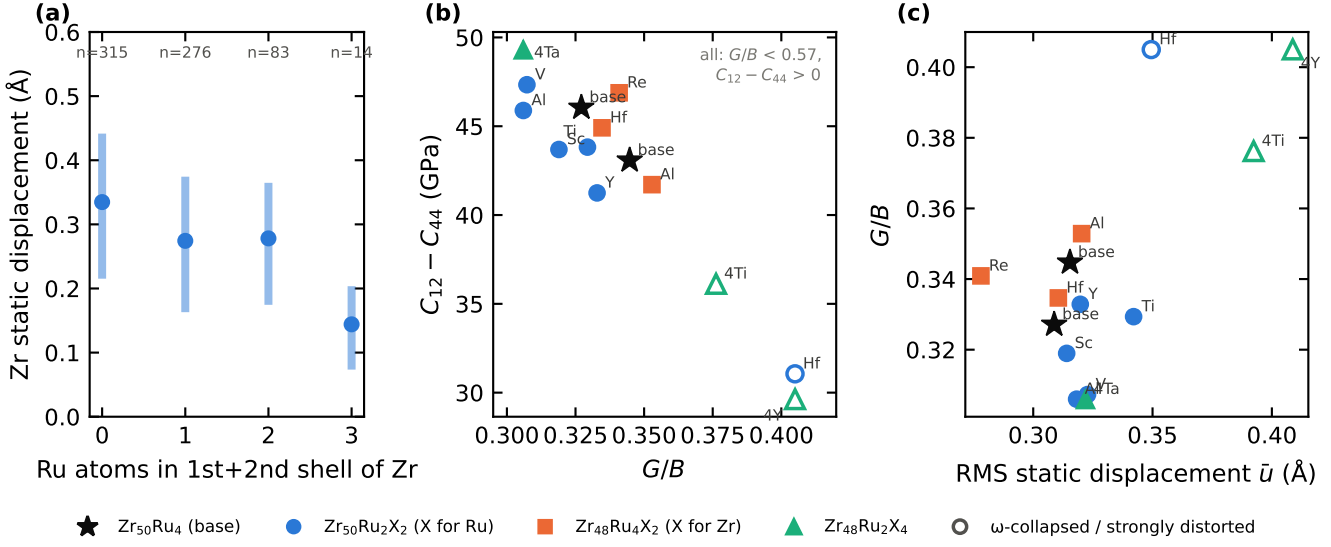
(a)
n=315
n=276
n=83
n=14
Zr static displacement (Å)
Ru atoms in 1st+2nd shell of Zr
(b)
all: G/B < 0.57,
C12 − C44 > 0
C12 − C44 (GPa)
G/B
4Ta
V
Al
Re
base
Hf
Ti
Sc
Y
4Ti
4Y
(c)
G/B
RMS static displacement ū (Å)
Hf
4Y
4Ti
Al
base
Re
Hf
Y
Ti
Sc
4Ta
Zr50Ru4 (base)
Zr50Ru2X2 (X for Ru)
Zr48Ru4X2 (X for Zr)
Zr48Ru2X4
ω-collapsed / strongly distorted